

MARKET RESEARCH & GO-TO MARKET STRATEGY FOR SPACE SATELLITE- SUBSYSTEMS FOR 'EARTH OBSERVATION (EO)' APPLICATION USING ZES'S SYSTEM-ON-MODULE 'ZSOM-GPU'

Fanni Djongkang
Yolanda Suciati
Jingyi Zhang
Wenxuan Zou





Outline

- 1 Introduction
- 2 Market Sizing: TAM & SAM
- 3 Market Analysis: Payload Segmentation
- 4 Competitive Landscape
- 5 Go-To Market Model

PURPOSE OF STUDY

Evaluation of commercial opportunities for Zero-Error Systems ZSOM-GPU radiation-tolerant computing platform in the Earth Observation (EO) sector

Mission Needs

Focus on commercial EO satellites requiring radiation-hardened semiconductor solutions for onboard processing capabilities

Regional Demand

Analyze GPU-based payload applications with focus on AI/edge computing demand in US and European markets.

Market Landscape

Examine EO Market Size (TAM&SAM), Growth Trends, and key satellite developers in commercial missions.

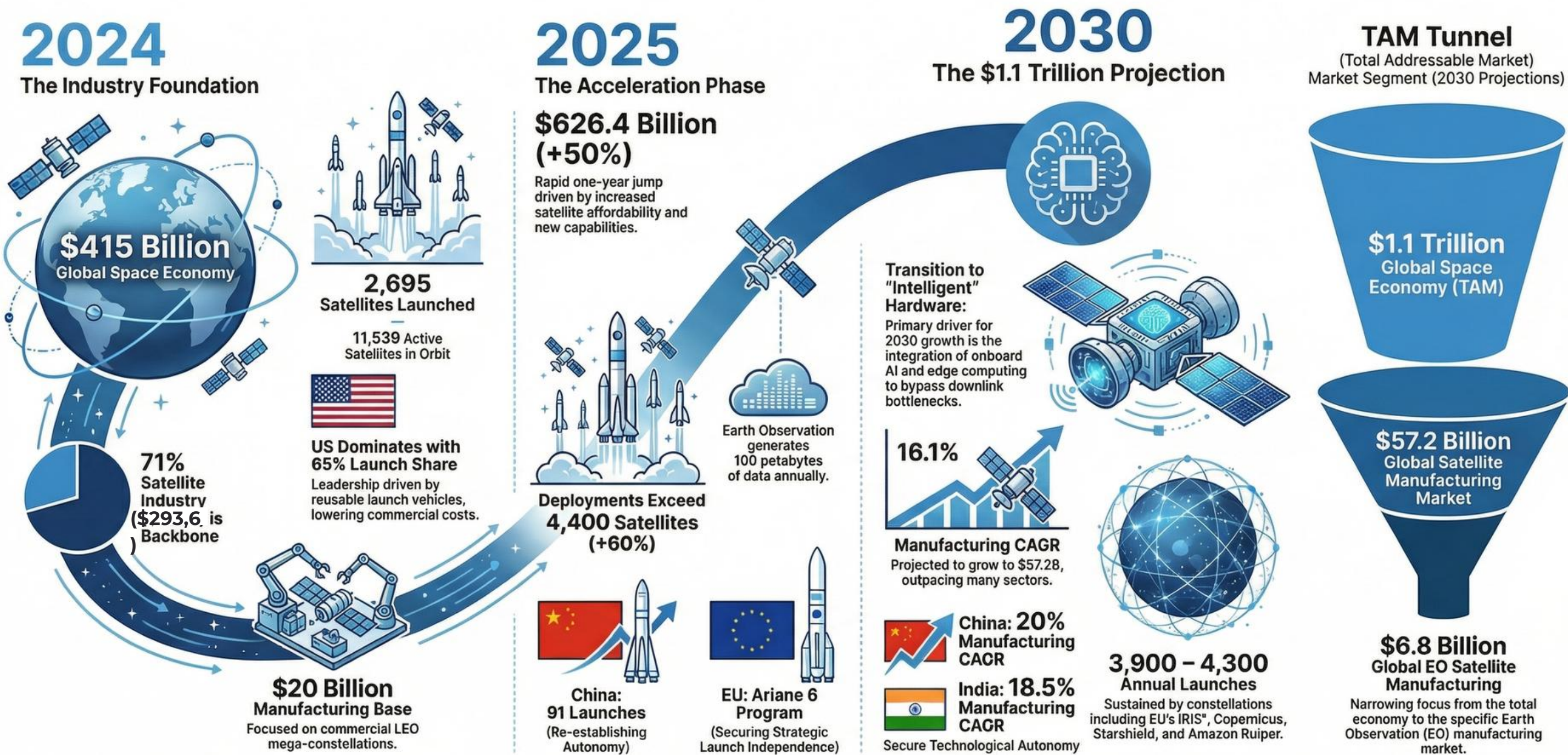
Strategic Roadmap

Develop Go-To-Market strategy targeting Tier-2 payload integrators and GPU suppliers with SWOT analysis of competitive positioning.



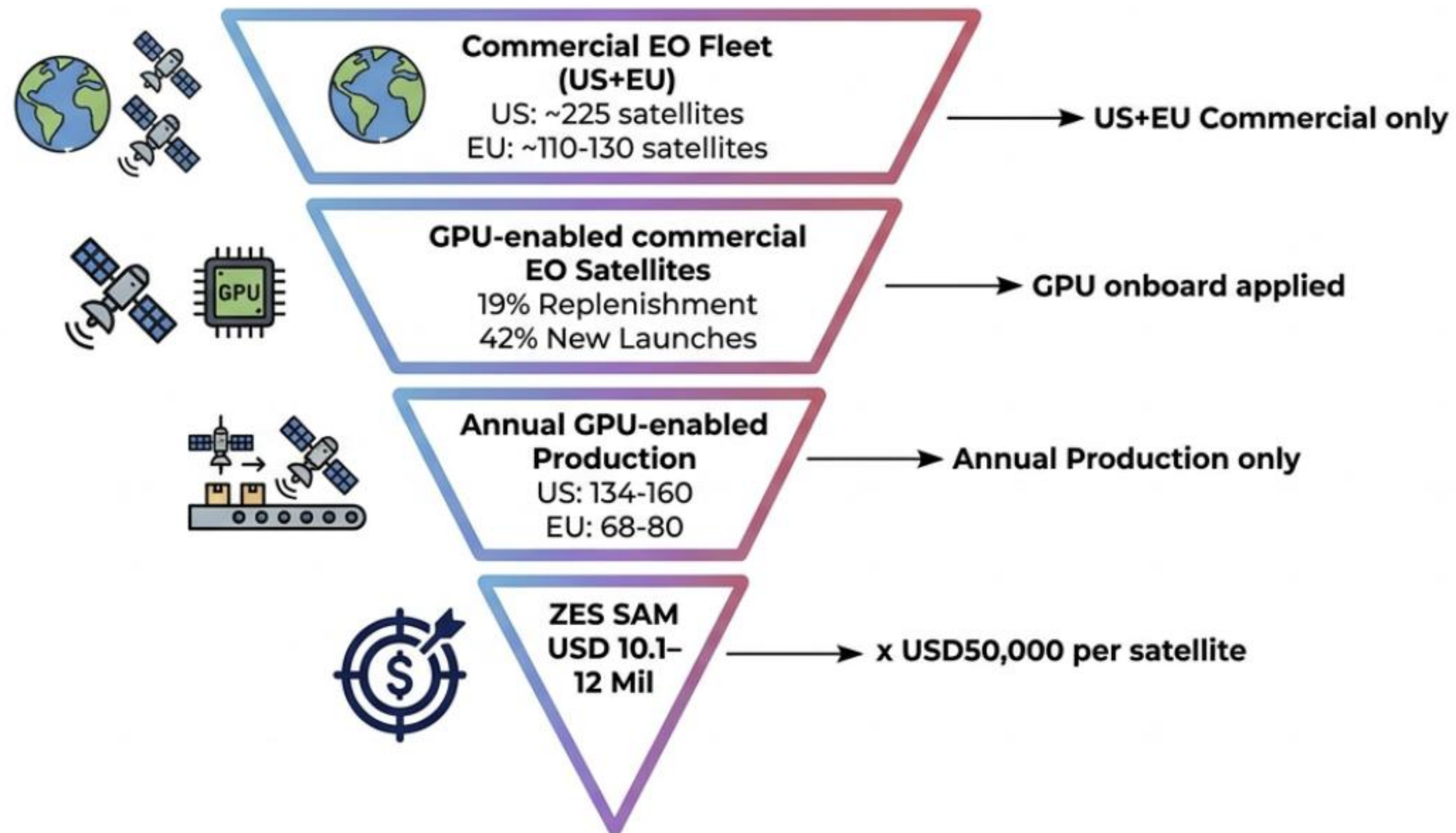
TOTAL ADDRESSABLE MARKET (TAM)

The Orbital Surge: Mapping the Global Space Economy (2024–2030)



SERVICEABLE ADDRESSABLE MARKET (SAM)

Target: **Commercial EO satellites** with **GPU-based** onboard processing in **US** and **Europe**, focusing on **Tier-2** payload integrators



GROWTH SCENARIO

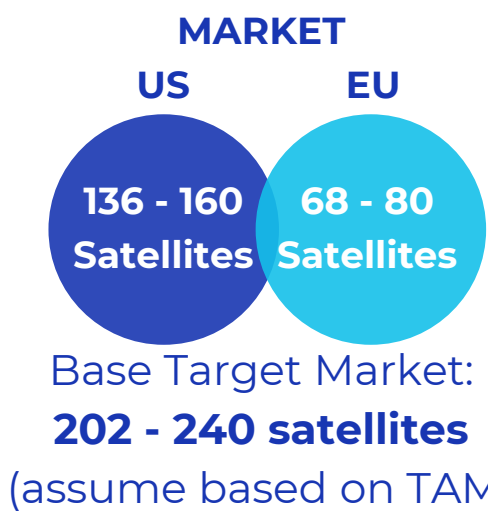


Latest shifts in GPU adoption are towards **Edge AI** and **Space-based Data Centres**.

Moreover, with NVIDIA's latest project to test GPUs as data centres in space (Pultarova, 2025), ambitious initiatives from several tech companies will increase GPU usage over the next couple of years.

Conservative Scenario

ZSOM-GPU as a high-performance “upgrade” tool.

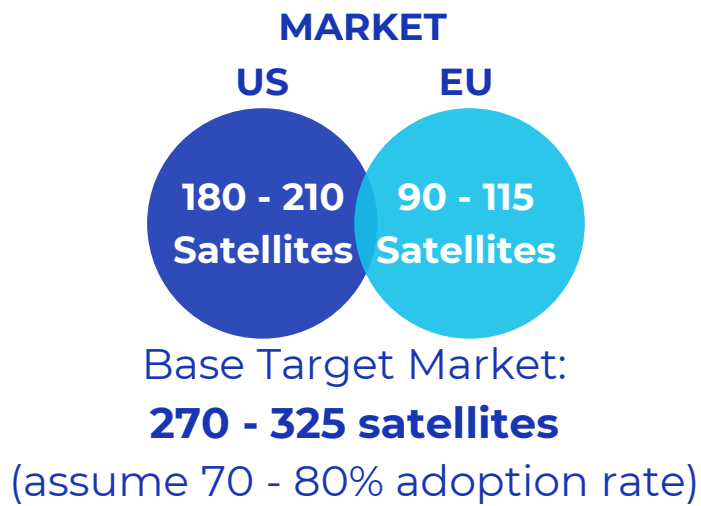


with **USD 50,000/unit**,
potential revenue:
USD 10.1 - 12M

- MARKET BARRIERS
-  **The Heritage Gap**
12 - 24 Month Flight Validation Requirement
 -  **Mission Longevity Risk**
5 - 7 Year Orbital Life for COTS Silicon
 -  **Critical Power Constraints**
Strict <20W Payload Power Envelope

Aggressive Scenario

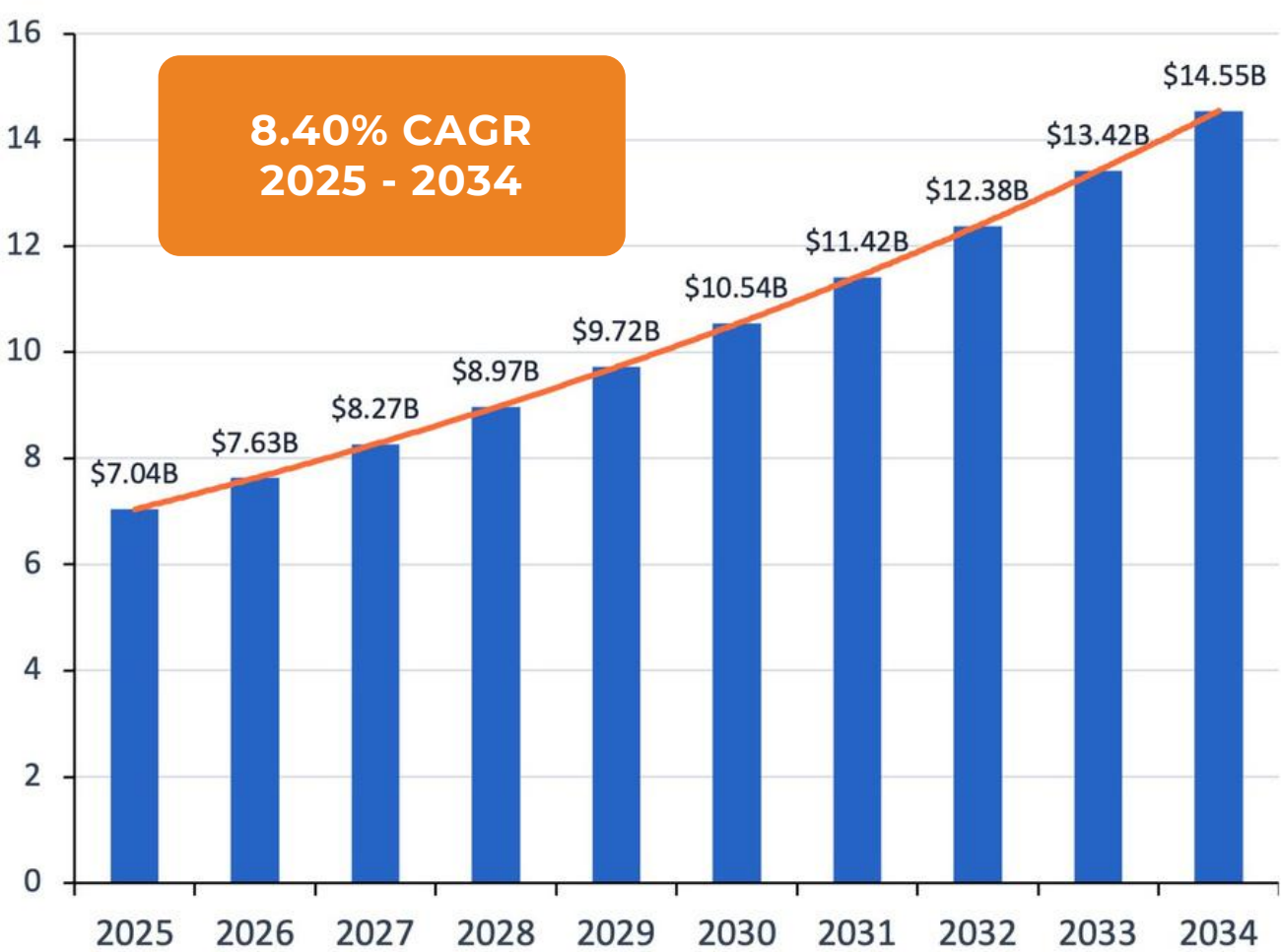
ZSOM-GPU as a default processor, to support industry’s pivot towards Full Edge AI Inference and real-time insights.



with **USD 60,000/unit**,
potential revenue:
USD 16.2 - 19.5M

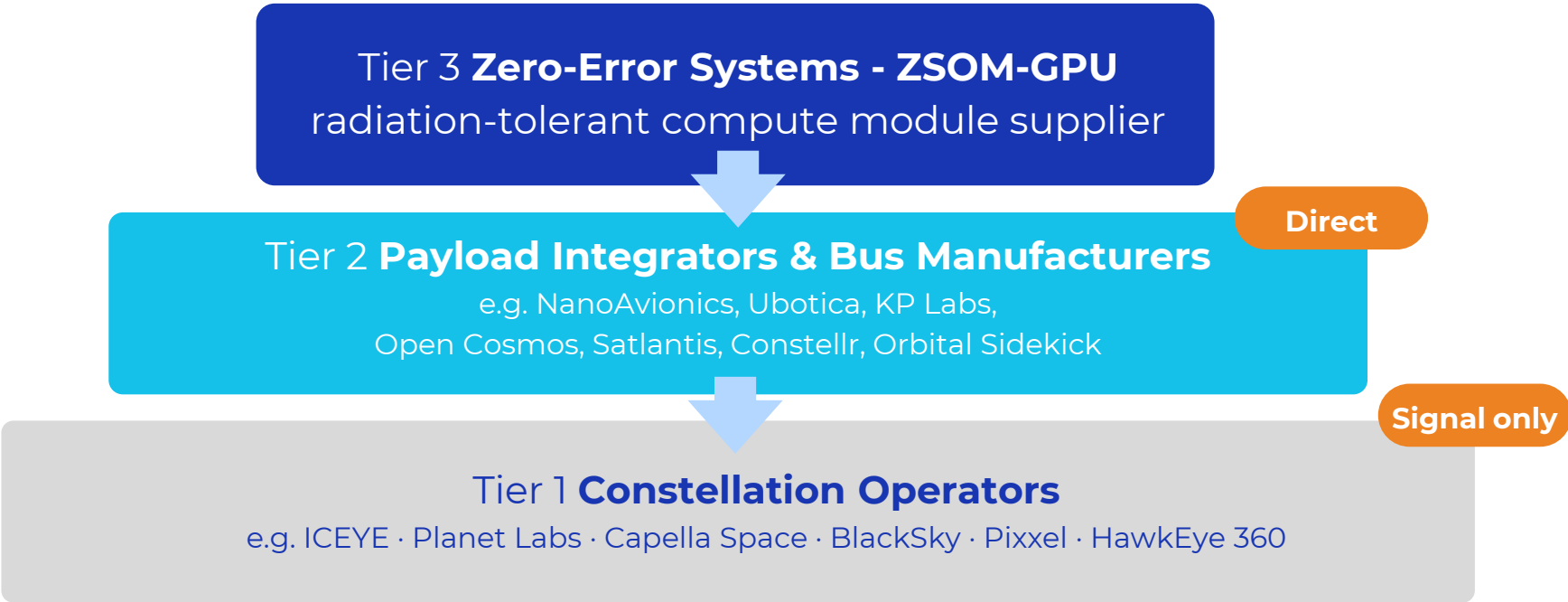
- MARKET BARRIERS
-  **Thermal Ceiling**
Managing 150W+ Heat Spikes within 20W Envelopes.
 -  **Connectivity Bottlenecks**
Upgrading Data-Buses for GPU-Speed Output.

Global EO Market



Onboard AI Value

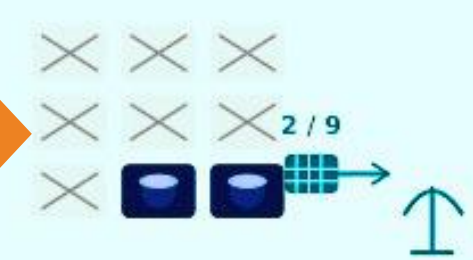
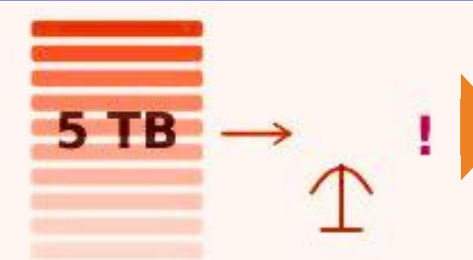
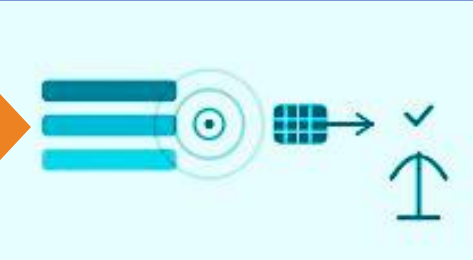
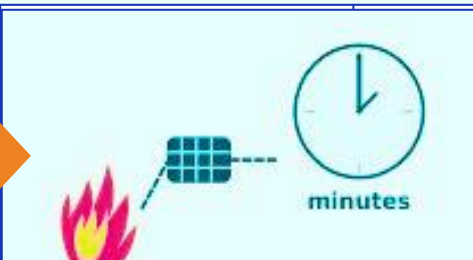
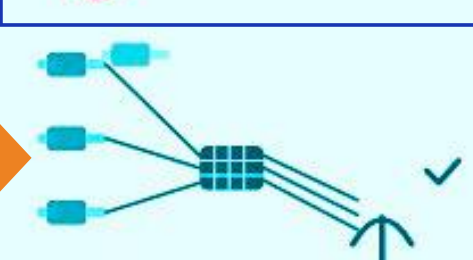
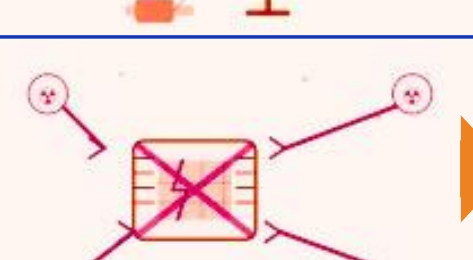
Onboard AI reduces transmitted data by **60 - 80%**, compressing time-to-insight from **hours to minutes**. Raw imagery pricing is a commodity; analytics subscriptions command premium margins.



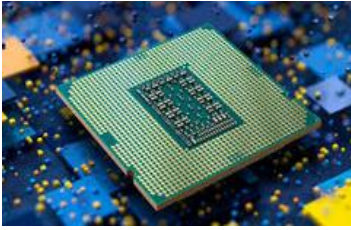
Supply Chain & Demand Signals

Tier 1	Scale	Contract	Signal to Tier 2
Planet Labs	200+ satellites in orbit	NRO 5-yr contract; edge compute delivers 80% data reduction	GPU compute is a spec requirement for its integrators
Capella Space	2 - 5 TB raw SAR per day	NRO SCE BAA Commercial Radar contract to mid 2027; \$15M US Air Force SAR acceleration contract	
BlackSky	3 Gen-3 sats, 12 by end 2026	35cm AI-enabled imagery; NGA GEOINT portal \$359M (2024)	NRO contract creates hard engineering deadlines for integrators
ICEYE	62 satellites, €250M in revenue 2025	Rheinmetall ICEYE €1.7B Germany contract to 2030 (options to €2.7B)	Integrators in supply chain have multi-year funded backlogs
HawkEye 360	36 sats, 12 formation clusters	NRO EUCOM mission contract Dec 2025; \$150M Series E equity and debt	Funded constellation expansion, active integrator procurement
Three-Body	12 launched May 2025; 2,800 target 2030	744 TOPS/satellite; 100 Gbps inter-satellite laser links	GPU-class orbital compute at scale is commercially viable

Market Size and Growth

Segment	2025 Market	Projected	By Year	GPU Relevance	GPU Solution		Tier 2 Target Integrators	Qualification Signal
Optical Imaging	\$3.1B	14.2% CAGR \$10.4B	2034	High			• Ubotica • Open Cosmos • Satlantis	Integrator's Tier 1 customers have onboard AI in spec.
SAR Imagery	\$1.59B	8.1% CAGR \$3.20B	2034	High			• NanoAvionics • Orbital Sidekick • KP Labs	FFT reconstruction is a spec item. ICD exists.
Thermal/Infrared Imaging	No publicly available market report isolates the commercial satellite thermal/IR EO segment (e.g. Constellr launched its first satellite in Jan 2025) Expected to emerge post-2027 with Copernicus LSTM (2028) and NASA SBG (2028) programmes.						• Constellr • Surrey ST • AAC Clyde	Tier 1 customer has compliance/regulatory mandate driving latency requirement.
RF/AIS Monitor	\$289M	7.2% CAGR \$505M	2032	Moderate			• HawkEye 360 supply chain • Spire	Constellation scale (>10 sats) creates onboard classification necessity and confirm batch plan.
Hosted AI Payloads	-	\$20B in ODC	2035	Long-term			• Axiom Space • OrbitsEdge • Starcloud	Funded ODC programme with thermal management solution defined.

CURRENT TECHNOLOGY ALTERNATIVES

Feature	CPU	FPGA	GPU
Image			
Best For	General Management	Specialized Logic	AI & Image Processing
Speed	Slow (Serial)	Fast (Fixed)	Extremely Fast (Parallel)
Effort	Low	Very High	Medium
Risk	Low	Low	High (Radiation)

Factor	FRONTGRADE	mercury	ZES ZERO-ERROR SYSTEMS
Market focus	Gov. space & system integrators	US DoD (85%+ revenue)	Commercial EO + gov. demos
Technology	Rad-tolerant storage & onboard CPU	FPGA/GPU-based compute	GPU-native AI architecture
AI/ML	Compute + storage + memory stack	FPGA signal + GPU boards	GPU-native, AI-optimized
Product Strategy	Modular Components	Intergrated Subsystems	Intergrated GPU Modules
Key Advantage	Heritage & Ecosystem	Prime Embed+ITAR lock	Space grade+ AI for big volume computing

COMPETITOR BENCHMARKING

WHY ZSOM WINS IN SPACE?

2200x
Fewer soft errors

60%
Cost Reduction

27
patents filed

5 - 10 yr
mission life



Advanced Radiation Hardening	Cost Advantage
ZSOM-GPU uses patented Latch-Up Protection and EDAC technology to achieve 2,200x fewer soft error rates than industry standards, ensuring that AI inference data remains completely accurate in high-radiation orbits.	By combining COTS commercial GPUs with ZES radiation protection technology, ZSOM-GPUs reduce cost to less than traditional space-scale chips (>\$50K) while supporting a mission life of 5 - 10 years.
Supply Chain - COTS	High Barrier Entry
Relying on the commercial supply chain of NVIDIA and AMD, ZSOM-GPUs can be delivered quickly in large quantities, and the ability to supply them in stock is a core advantage that traditional space manufacturers cannot replicate.	ZES' patented LDAP-IC technology creates a high barrier to competition, making it difficult for competitors to achieve equal reliability in space radiation environments without the patent.

SWOT ANALYSIS

INTERNAL CAPABILITIES

STRENGTHS

- ST 1 Patented and Exclusive Radiation-Hardening Technology
- ST 2 First-to-Market COTS-based SOMs
- ST 3 COTS Cost Advantage Over Traditional Rad-Hard Alternatives
- ST 4 Credibility through institutional backing (e.g. Airbus Ventures) and technical IP
- ST 5 Airbus Ventures and Ecosystem Access
- ST 6 Deep Academic and Semiconductor Expertise
- ST 7 Singapore as a Strategic Operating Base
- ST 8 Industrial Shield for NVIDIA AI

WEAKNESSES

- WS 1 Market Entry Lead Time for GPU Modules
- WS 2 Early-Stage Flight Qualification Cycle
- WS 3 Small Company Scale Relative to the Sales Cycle
- WS 4 Quality Assurance for Batch Integration
- WS 5 Revenue and Funding Concentration Risk
- WS 6 Fragmented Global Brand Recognition

EXTERNAL ENVIRONMENT

OPPORTUNITIES

- OP 1 Surge in GPU – Driven EO
- OP 2 High-Leverage Entry via Tier-2 Integrators
- OP 3 European “Sovereignty Premium” and Defence Growth
- OP 4 Policy-Driven Demand for Thermal IR Monitoring
- OP 5 National Space Agency of Singapore (NSAS) Catalyst
- OP 6 First Mover IP in High-Performance GPU Protection

THREATS

- TS 1 Market Dominance by Well-Capitalised Incumbents
- TS 2 Structural Entry Barrier of the “Experience Gap”
- TS 3 Thermal Limitation on Orbital Performance
- TS 4 ITAR/Export Friction in the US Defence Market
- TS 5 Rising Competition from State-Backed Chinese Networks
- TS 6 Accelerated Scaling of Established AI Rivals
- TS 7 Deep-Tech Funding and Execution Risks



GO-TO-MARKET CANVAS

Products

- Radiation-tolerant COTS-based GPU compute module for EO satellites
- Enables onboard AI inference, image preprocessing, object detection, and analytics
- Built from Commercial-Off-the-Shelf (COTS) based

Target Market

- Anchor Revenue (Tier 2 integrators & bus manufacturers): NanoAvionics, Open Cosmos,
- High/Strategic Margin (Payload Developers): KP Labs, Ubotica, Satlantis, Orbital Sidekick, Constellr, Capella Space
- Expansion Market (Direct Constellation Operator): Planet Labs, ICEYE, Pixxel
- Geographic focus: North America & Europe have ~60% of active commercial EO satellites

Distribution

- Direct engagement with satellite manufacturers, payload developers & constellation operators
- Positioned as a component within the broader satellite supply chain

Value Proposition

Performance

- Radiation-tolerant COTS GPUs: 60 - 80% data volume reduction, hours-to-minutes insight, higher AI throughput than FPGAs

Radiation Hardening

- 2,000 - 2,200× fewer soft errors than FPGA SECEDED; latch-up detection with auto self-reset; multi-year LEO reliability verified

Cost & Supply Chain

- Below \$50K unit cost, 5 - 10 year mission life; NVIDIA/AMD stock availability; CUDA/PyTorch/TensorFlow compatible

IP Moat

- 27 patents (6 granted) on LDAP-IC that is unreplicable without infringing ZES IP

Launch Strategy

- Platform-level integration (e.g. NanoAvionics with 300+ backlog)
- Performance upgrade → SmallSat → default onboard AI processor
- Conservative: efficiency tool/Aggressive: default AI inference
- NVIDIA ecosystem + Starcloud, Axiom Space (post heritage)
- Tier 2 integrators, the primary GTM, US & EU coverage

Competitors

- Frontgrade & Mercury dominate with radiation-hardened CPU/FPGA solutions, strong heritage, but not optimised for AI or high-performance EO processing.

Assumption/Risks

- EO missions are rapidly adopting onboard AI and high-performance computing
- Tier-2 integrators provide the fastest path to market entry
- European defense spending and sovereignty drive procurement demand
- Shift toward cost-efficient COTS solutions over traditional rad-hard systems
- Long qualification cycles delay revenue realization
- Strong incumbents and AI players outcompete and scale faster
- Thermal and in-orbit constraints threaten reliability and adoption
- Deep-tech funding and scaling challenges limit growth

Considerations

- Overcome the “heritage gap” through partnerships with Tier-2 integrators
- Ensure reliability under radiation, thermal, and power constraints in space
- Position ZSOM-GPU as a complementary enabler (“industrial shield”) rather than a direct replacement
- Build institutional credibility and global brand recognition
- Achieve scalable quality assurance and production readiness
- Navigate regulatory and export control barriers (e.g., ITAR)

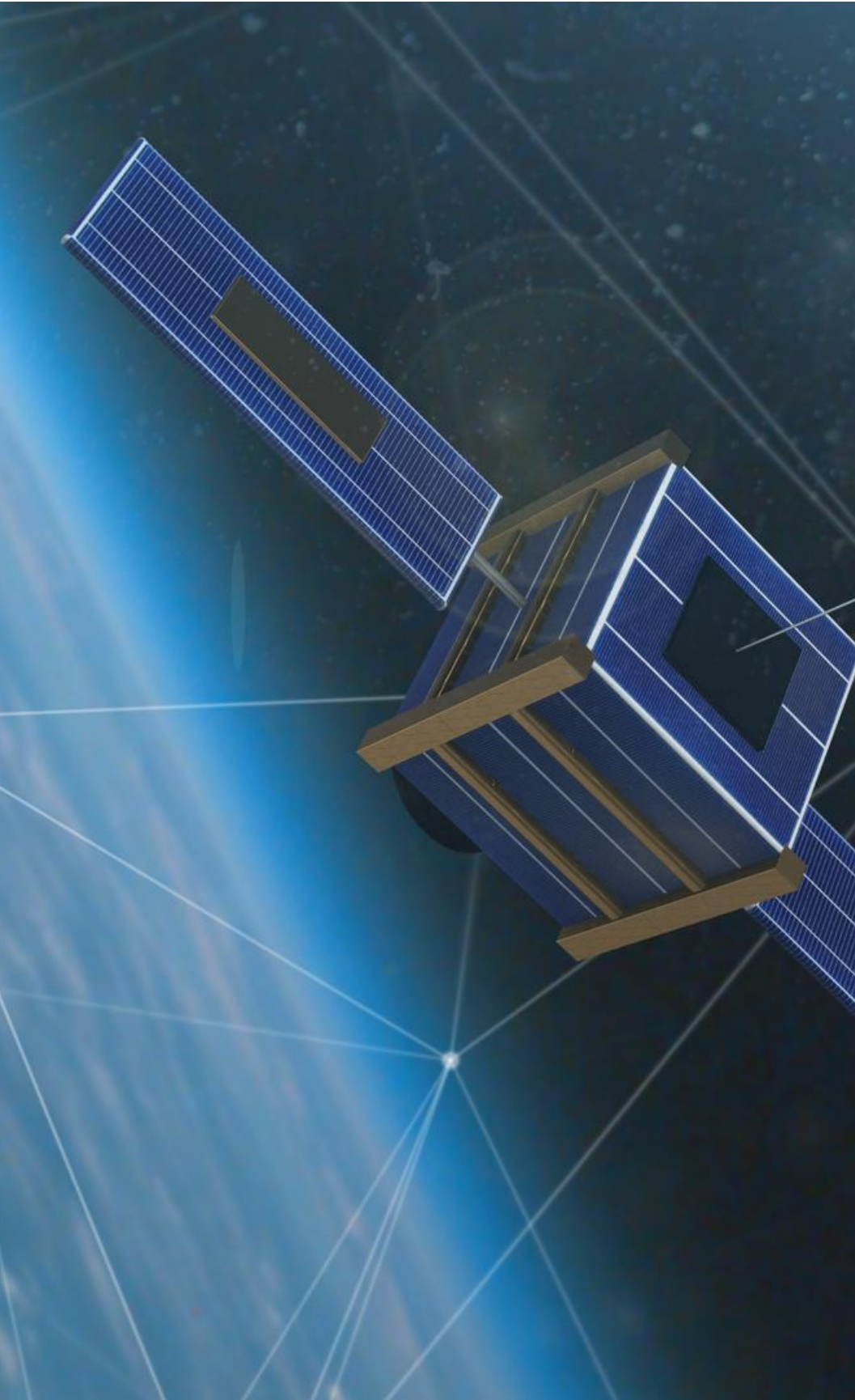
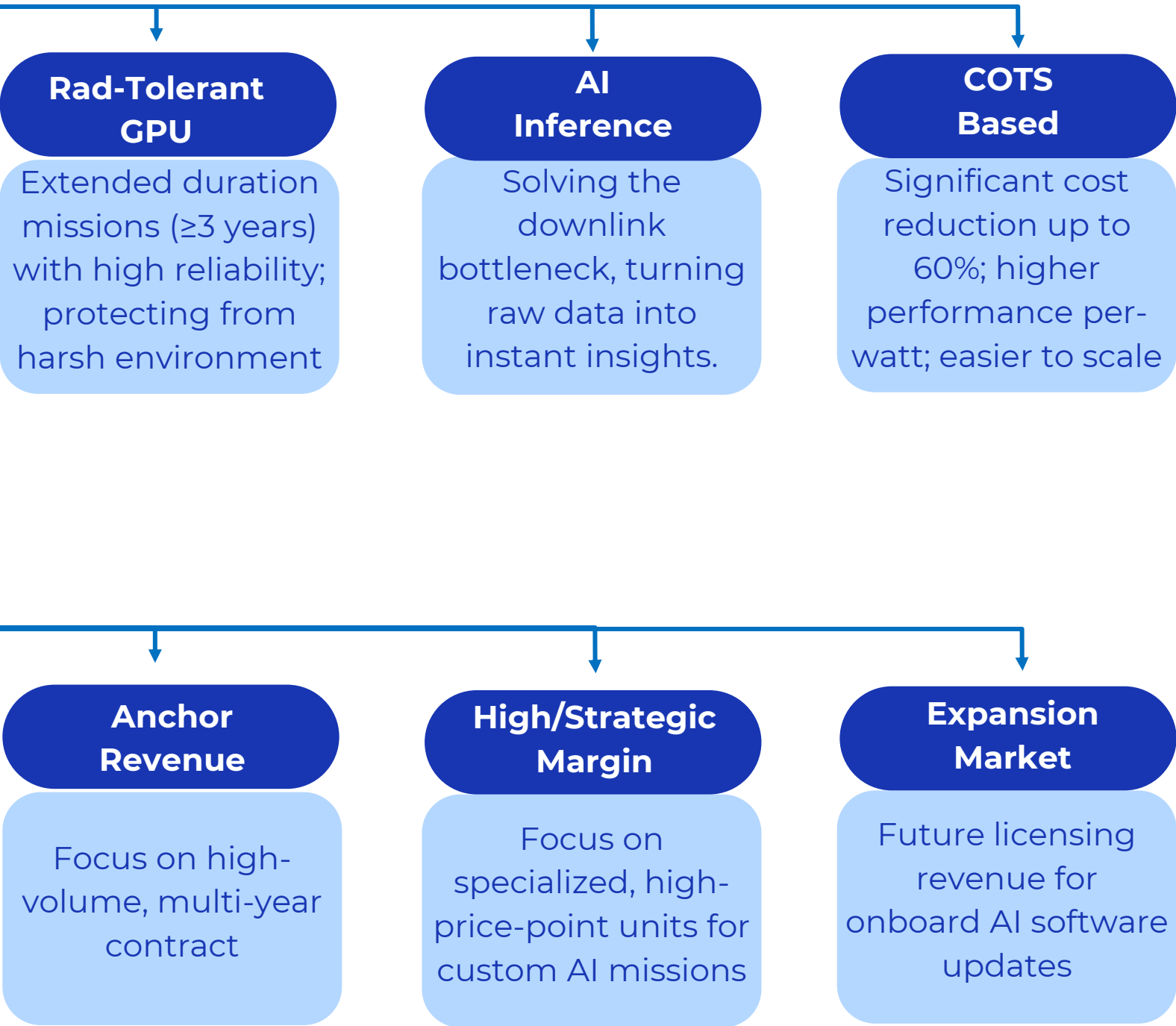
Products

- Radiation-tolerant GPU compute module for EO satellites
- Enables onboard AI inference, image preprocessing, object detection, and analytics
- Built from Commercial-Off-the-Shelf (COTS) based

Target Market

- Anchor Revenue (Tier 2 integrators & bus manufacturers): NanoAvionics, Open Cosmos,
- High/Strategic Margin (Payload Developers): KP Labs, Ubotica, Satlantis, Orbital Sidekick, Constellr, Capella Space
- Expansion Market (Direct Constellation Operator): Planet Labs, ICEYE, Pixxel
- Geographic focus: North America & Europe are ~60% of active commercial EO satellites

Strategic Benefit



VALUE PROPOSITION

Performance

- COTS GPUs bridging commercial performance and space qualification
- 60 - 80% data volume reduction, cutting bandwidth costs
- Time-to-insight: hours → minutes
- Higher AI throughput than FPGAs, no custom firmware needed

Radiation Hardening

- Auto self-reset on particle hit
- 2,000 - 2,200× fewer soft errors than FPGA SECEDED
- Verified: 14 bit-flips in 29 KB image fully recovered
- Multi-year LEO reliability for AI inference and scientific data

Cost & Supply Chain

- Unit cost below \$50K, 5 - 10 year mission life
- NVIDIA/AMD supply chain - available from stock, high-volume scalable
- Compatible with CUDA, PyTorch, and TensorFlow

IP Moat

27 patents (6 granted) on LDAP-IC that competitors cannot replicate without infringing ZES IP

Competitors

Frontgrade & Mercury dominate with strong heritage but CPU/FPGA architectures cannot scale with AI demand. ZES should target the commercial EO integrators they cannot serve and position GPU-native AI as the upgrade path they cannot offer.

Launch Strategy & Distribution

ZES Current Position

- Strong IP
- Limited flight heritage
- Small commercial team

Direct sales may not scale

- *12 - 24 month hardware sales cycle*
- *Depends on prior qualification history*
- *Requires sustained in-person engagement that ZES may not be able to maintain at scale*

Partnership can...

- Embed ZSOM-GPU in partner's existing customer backlog
- Accumulate *flight hours* across multiple missions simultaneously
- Cover *geographic + regulatory* constraints: US defence, EU sovereign

BMP5300 Project #28

Distribution

SIGNED

Domestic supply chain routing and ITAR compliance for US government and defence procurement. Handles logistics.

Market access & compliance (not revenue generation) - conservative approach

Platform Integration

PRIORITIZE

ZSOM-GPU included in partner's standard hardware stack across all programs of a given class.

Primary GTM motion -> revenue, flight heritage, scale

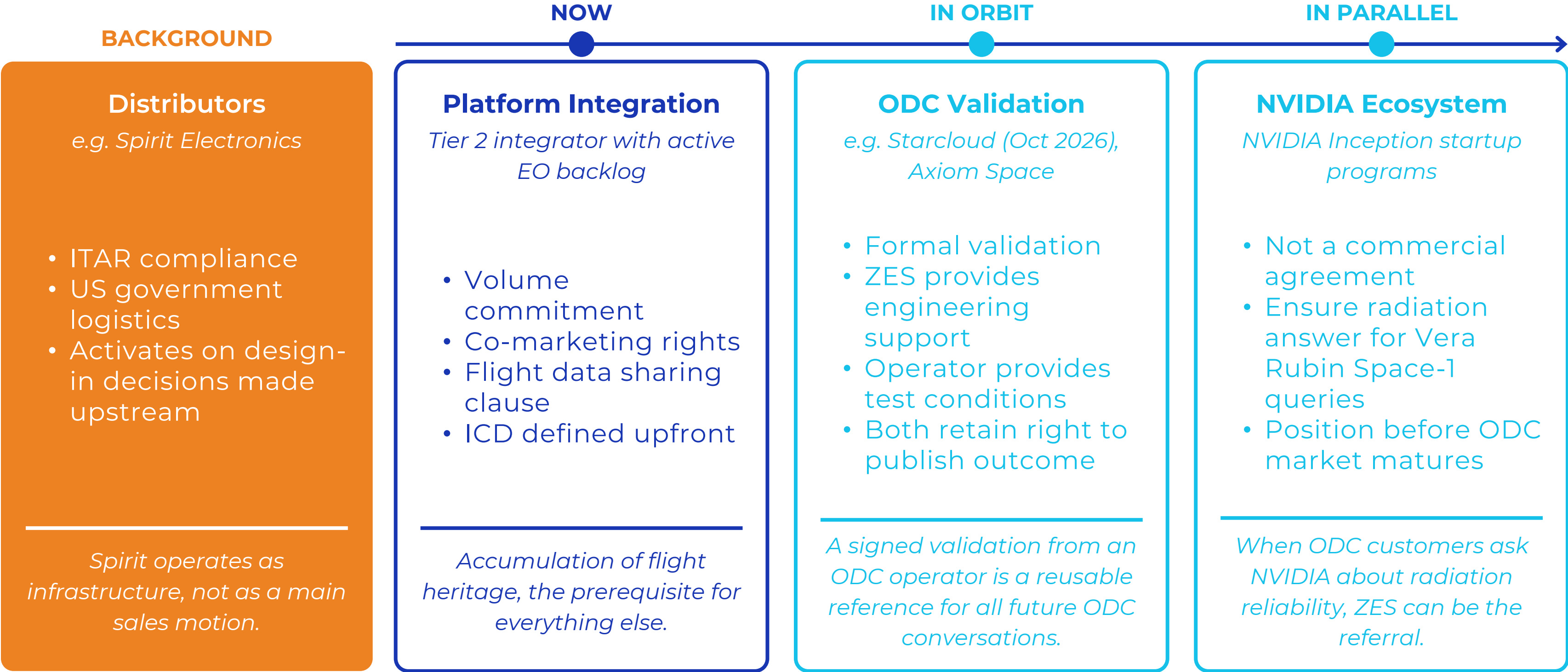
ODC Operators

PIPELINE

Space-based platforms processing data in orbit. Both face the radiation reliability problem that ZES solves. Starcloud launches Oct 2026 with NVIDIA H100/Blackwell. Axiom deployed 2 ODC nodes early 2026.

Bull-case suggests \$20B market, conditional on radiation-tolerant processors

PARTNERSHIP SEQUENCING



ASSUMPTIONS

EO missions are rapidly adopting onboard AI and high-performance computing

Tier-2 integrators provide the fastest path to market entry

European defense spending and sovereignty drive procurement demand

Shift toward cost-efficient COTS solutions over traditional rad-hard systems

RISKS

Long qualification cycles delay revenue realization

Strong incumbents and AI players outcompete and scale faster

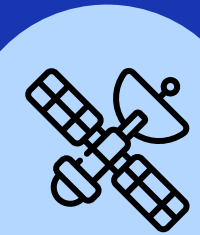
Thermal and in-orbit constraints threaten reliability and adoption

Deep-tech funding and scaling challenges limit growth

CONSIDERATIONS



Overcome the “heritage gap” through partnerships with Tier-2 integrator



Ensure reliability under radiation, thermal, and power constraints in space

Position ZSOM-GPU as a complementary enabler (“industrial shield”) rather than a direct replacement



Build institutional credibility and global brand recognition

Achieve scalable quality assurance and production readiness



Navigate regulatory and export control barriers (e.g ITAR)

THANK YOU!